

WEEK 2 TASK: ADVANCED DATA VISUALIZATION AND STORYTELLING WITH PYTHON

Project Title: Milk Quality Analysis – Advanced Data Visualization and Storytelling

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Domain: Data Science

Tools Used: Python, Pandas, Matplotlib, Seaborn, Google Colab

1. Introduction

Data visualization is an important part of Data Science because it converts numerical data into understandable visual patterns. In this task, the Milk Quality Dataset was analyzed using different visualization techniques to identify trends, relationships, variations, and patterns in milk quality.

2. Objectives

1. To create meaningful visualizations using Python.
2. To identify trends and relationships in the dataset.
3. To understand milk quality patterns.
4. To communicate insights to non-technical users.
5. To identify business and scientific implications from the analysis.

3. Dataset

The Milk Quality Dataset contains pH, Temperature, Taste, Odor, Fat, Turbidity, Colour, and Grade. The first seven variables represent milk characteristics, while Grade represents the overall milk quality classification.

4. Methodology

The analysis followed the stages of data loading, data preparation, visualization development, interpretation, and data storytelling. Python libraries including Pandas, Matplotlib, and Seaborn were used to create and analyze the visualizations.

5. Visualization Analysis

5.1 Distribution of Numerical Parameters

Histograms were used to understand the distribution and spread of numerical milk quality parameters. They help identify common ranges, concentration of observations, and unusual values.

5.2 Correlation Heatmap

A correlation heatmap was used to examine relationships between numerical milk quality parameters. It helps identify variables that show stronger positive or negative relationships.

5.3 Milk Quality Grade Distribution

A bar/count plot was used to compare the number of observations belonging to different milk quality grades. This provides an understanding of the distribution of the target variable.

5.4 Box Plot Analysis

Box plots were used to examine the median, spread, variability, and possible outliers in numerical milk quality parameters.

5.5 Relationship Analysis

A relationship-based visualization was used to examine the association between selected milk quality parameters. This helps identify patterns that can be investigated further using statistical analysis and machine learning.

6. Data Storytelling

The visual analysis begins with understanding the distribution of milk quality parameters. It then examines relationships between variables, the distribution of quality grades, variability, and selected parameter relationships. Together, these visualizations provide a clear and meaningful story about the characteristics of the milk quality dataset.

7. Key Insights

1. Milk quality can be studied using multiple physical and sensory characteristics.
2. Numerical parameters show different distributions and levels of variability.

3. Correlation analysis helps identify relationships between milk quality parameters.
4. Grade distribution provides information about the overall quality categories.
5. Visualization helps identify unusual observations and important patterns.
6. The findings provide a foundation for statistical hypothesis testing and machine learning.
8. Business and Scientific Implications

The analysis can support systematic milk quality monitoring and data-driven decision-making. It can help identify unusual samples, support quality-control processes, and provide a foundation for automated milk quality assessment. The relationships observed in the visual analysis can also be investigated further using statistical and machine learning techniques.

9. Conclusion

The Week 2 task successfully demonstrated advanced data visualization and storytelling using the Milk Quality Dataset. Different visualization techniques were used to understand distributions, relationships, quality grades, and variability. The analysis provided useful insights into milk quality and created a foundation for further statistical analysis and machine learning.

10. Code and Visualization Outputs

The complete Python code used for the analysis and visualization, along with all generated visualization outputs, is provided in the attached PDF.

Appendix A: Complete Python Code and Visualization Outputs

11. References

1. Python Documentation.
2. Pandas Documentation.
3. Matplotlib Documentation.
4. Seaborn Documentation.
5. Milk Quality Dataset.

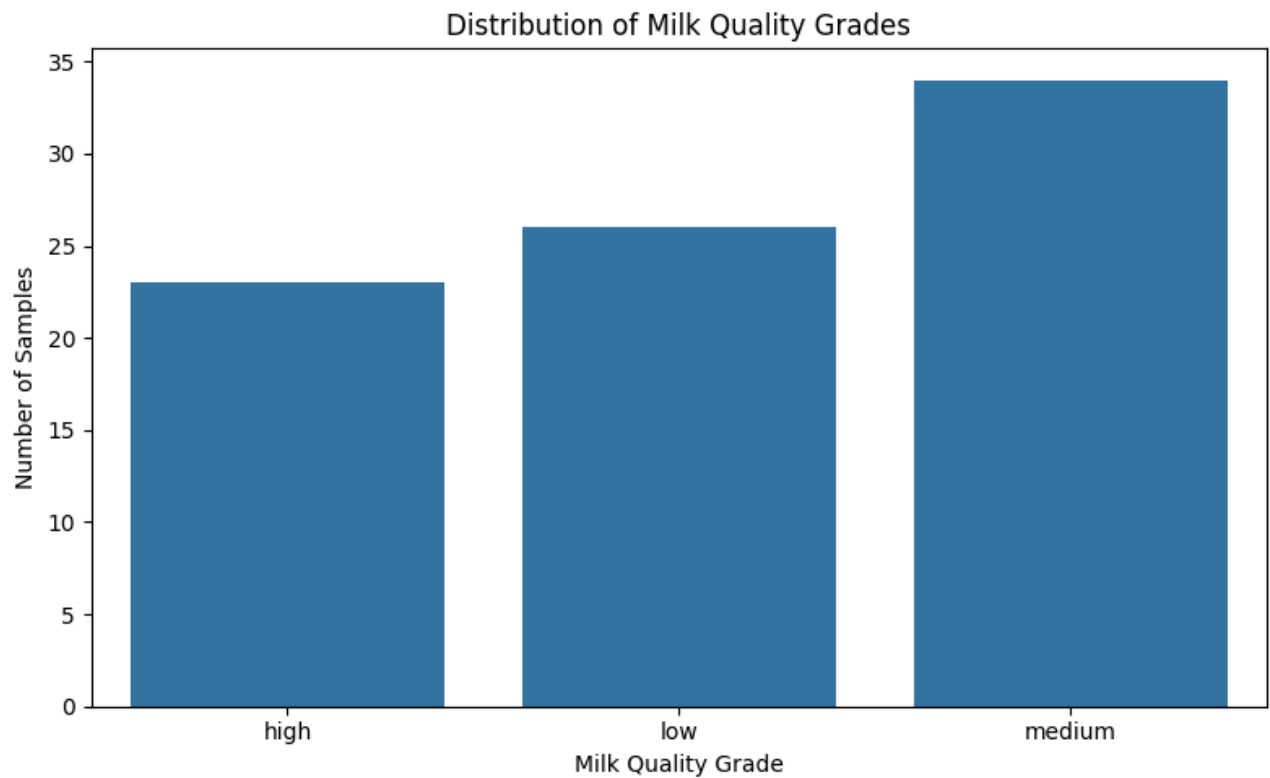
```
In [18... import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 5))

sns.countplot(data=df, x='Grade')

plt.title('Distribution of Milk Quality Grades')
plt.xlabel('Milk Quality Grade')
plt.ylabel('Number of Samples')

plt.tight_layout()
plt.show()
```

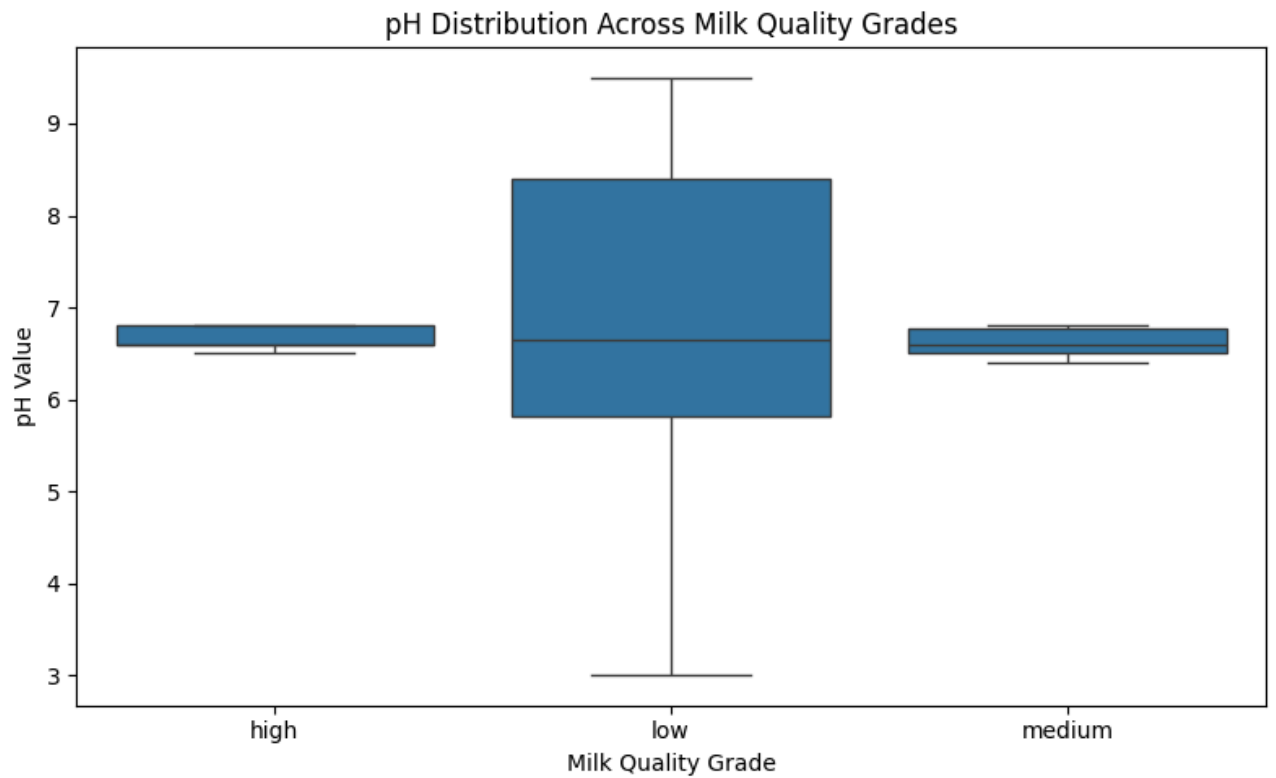


```
In [19... plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x='Grade', y='pH')

plt.title('pH Distribution Across Milk Quality Grades')
plt.xlabel('Milk Quality Grade')
plt.ylabel('pH Value')

plt.tight_layout()
plt.show()
```

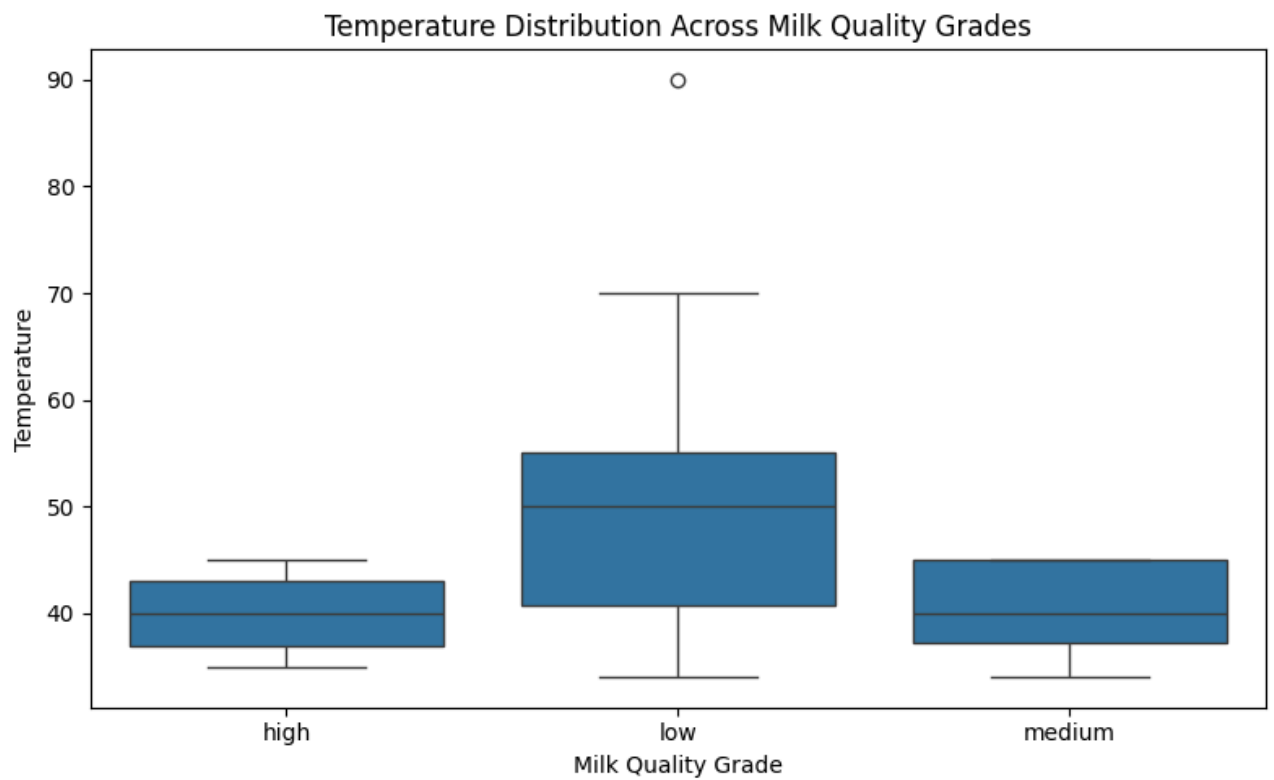


```
In [20... plt.figure(figsize=(8, 5))

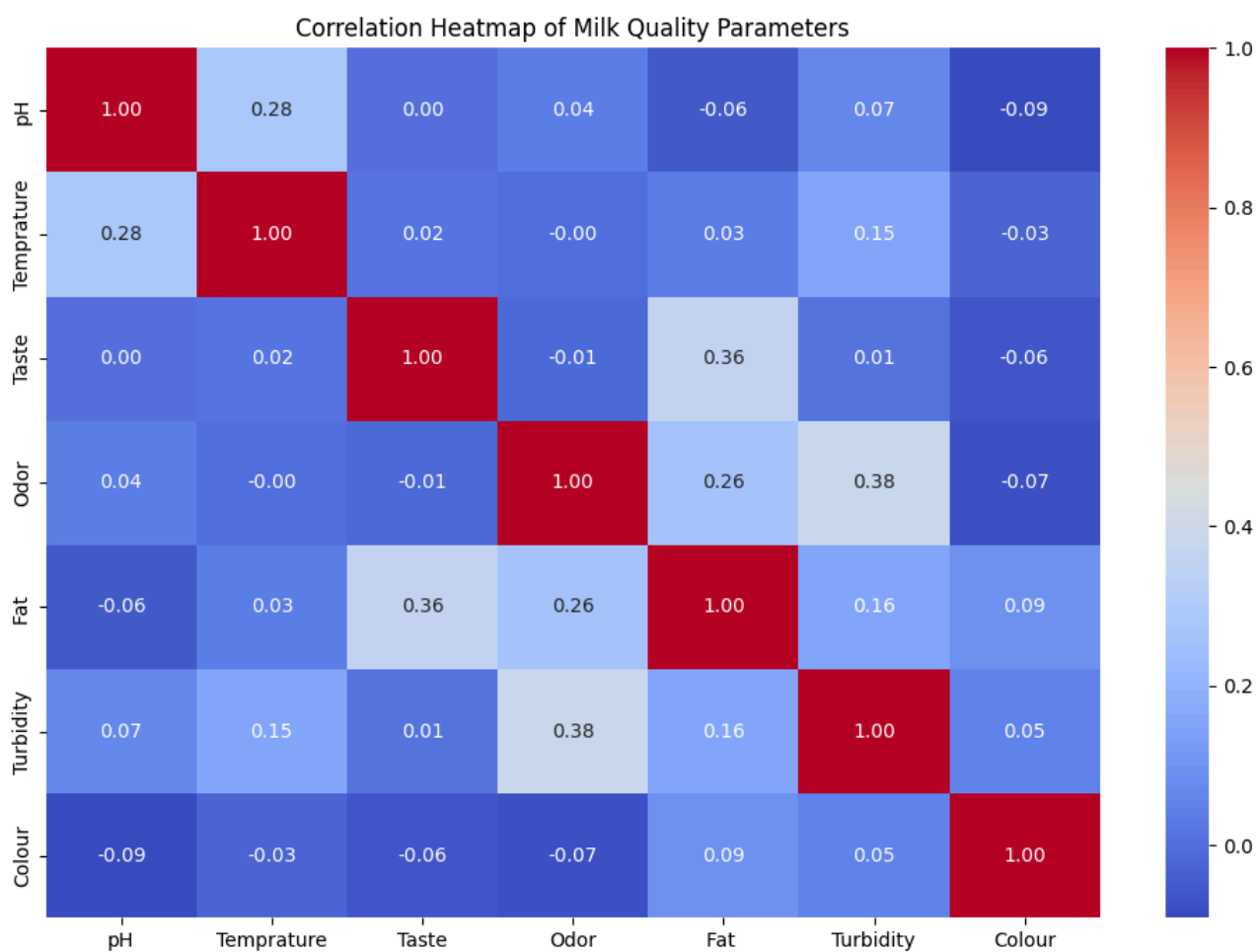
sns.boxplot(data=df, x='Grade', y='Temperature')

plt.title('Temperature Distribution Across Milk Quality Grades')
plt.xlabel('Milk Quality Grade')
plt.ylabel('Temperature')

plt.tight_layout()
plt.show()
```



```
In [21... plt.figure(figsize=(10, 7))  
  
correlation = df.select_dtypes(include='number').corr()  
  
sns.heatmap(correlation, annot=True, cmap='coolwarm', fmt='.2f')  
  
plt.title('Correlation Heatmap of Milk Quality Parameters')  
  
plt.tight_layout()  
plt.show()
```



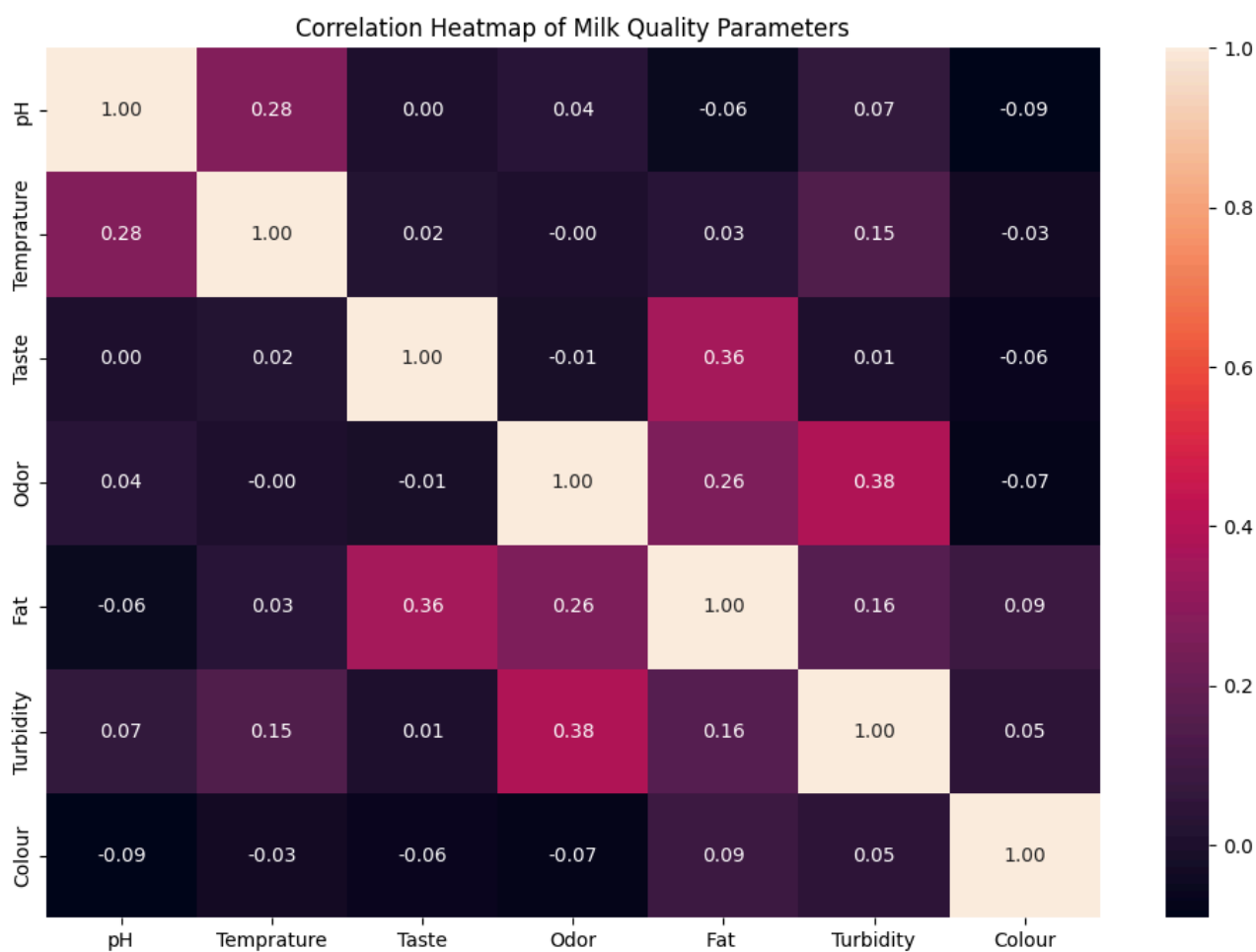
```
In [26... import matplotlib.pyplot as plt
import seaborn as sns

num_df = df.select_dtypes(include='number')

correlation = num_df.corr()

plt.figure(figsize=(10, 7))
sns.heatmap(correlation, annot=True, fmt='.2f')

plt.title('Correlation Heatmap of Milk Quality Parameters')
plt.tight_layout()
plt.show()
```



```
In [27... import seaborn as sns
import matplotlib.pyplot as plt

# Select all numerical columns
numeric_columns = df.select_dtypes(include='number').columns

# Create pairplot
sns.pairplot(df, vars=numeric_columns, hue='Grade')

plt.suptitle('Pairwise Relationships Among Milk Quality Parameters', y=1.
plt.show()
```

Pairwise Relationships Among Milk Quality Parameters

